



Charlie (Zhaohua) Qi

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Portfolio: charlie-qi394.github.io/charlie-qi-portfolio

PROFESSIONAL SUMMARY

Master of Computer Science graduate specialising in Artificial Intelligence, targeting software engineering, Python backend, data, automation and AI-adjacent roles. Built public projects across React/FastAPI product workflows, FastAPI/Streamlit applications, PostgreSQL/pgvector, OpenAI embeddings, LangGraph RAG workflows, machine learning, NLP, computer vision, graph algorithms and applied cybersecurity. Brings 10 years of experience across biotech/nutrition product R&D, infant formula manufacturing and healthcare nutrition, with transferable strengths in automation, data handling, business systems, process improvement, technical documentation, stakeholder support and practical problem solving. Demonstrates learning agility and commitment to career transition through first-in-class results in three computer science units and a High Distinction average.

TECHNICAL SKILLS

Software / Backend

Python, FastAPI, Streamlit, REST APIs, React/Tailwind exposure, pytest, Docker Compose

AI / RAG

OpenAI embeddings, Gemini scanning exposure, vector search, LangGraph, citations

ML / NLP / CV

PyTorch, TensorFlow/Keras, scikit-learn, Seq2Seq, CNNs

Data

SQL, PostgreSQL, pgvector, SQLAlchemy/SQLite exposure, pandas, data preprocessing

Automation

VBA tools, HTML calculator, SAP/M3 exposure, process improvement

Security

RSA, AES, X.509, CSRs, CRLs, PKI concepts

SOFTWARE, DATA AND AI PROJECTS

FridgePeace Shared-Household Food Management PWA

[GitHub](#)

React, FastAPI, AI scanning workflow

- Contributed to a public portfolio snapshot of a shared-household food-management PWA with React/Tailwind frontend, FastAPI/SQLAlchemy/SQLite backend, pantry workflows, expiry tracking and Gemini-backed AI scanning support.
- Supported user research/problem framing, epics, user stories, acceptance-criteria traceability, AI scanning workflow handoff, testing evidence structure and project documentation cleanup for a team software-engineering project.

AI Regulatory Knowledge Assistant

[GitHub](#)

RAG, FastAPI, PostgreSQL/pgvector

- Built a document-grounded RAG assistant using FastAPI, Streamlit, PostgreSQL/pgvector, OpenAI embeddings and chat generation to answer questions over local regulatory documents with citations and query history.
- Implemented TXT/PDF/DOCX ingestion, chunking, vector retrieval, LangGraph orchestration, Docker Compose setup and simple CSV plus optional RAGAS evaluation workflows.

Python PKI Certificate System

[GitHub](#)

Python, cryptography, X.509

- Implemented an educational PKI simulator with Root CA, Sub-CAs, clients, encrypted certificate requests, validation and revocation.

Seq2Seq Recipe Generation with Attention

[GitHub](#)

NLP, PyTorch

- Built a PyTorch LSTM encoder-decoder recipe-generation system using ingredient lists as source sequences and recipe instructions as targets.

Computer Vision Classification and Segmentation

[GitHub](#)

CV, TensorFlow/Keras

- Implemented Harris corner detection and Canny edge detection, comparing custom outputs with library baselines.

AI Graph Search Algorithms

[GitHub](#)

Python, algorithms, CLI

- Implemented BFS, DFS, Greedy Best-First Search and A* over weighted CSV graph inputs with a reusable package API, command-line interface and focused tests.

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Biotech product R&D, manufacturing, healthcare and transferable systems experience

BIOTECH PRODUCT R&D, MANUFACTURING AND SYSTEMS

Technical Manager - Research and Development in Infant Formula

Melbourne | Apr 2022 - Present

Bellamy's Organic

- Lead infant formula and nutritional product development from concept to launch, coordinating marketing, sales, suppliers, regulatory stakeholders and third-party manufacturers.
- Manage formulation assessment, nutrition panels, ingredient sourcing, packaging validation, sensory evaluation, shelf-life validation and controlled technical documentation.
- Support regulatory and market-access work including SAMR GB-2 registration, China organic WIT certification and FDA exemption-related technical evidence.
- Use SAP and cross-functional data inputs to support technical decisions, supplier coordination, product changes and operational follow-up.
- Developed an HTML ingredient forecasting calculator to support sales and procurement planning, improving business visibility and stakeholder alignment.

Formulation Technologist - Operation

Jan 2020 - Apr 2022

Fonterra Australia

- Supported new product/process development, drying and processing challenge resolution, equipment commissioning and campaign-management improvement in a manufacturing environment.
- Used SAP and production data to support formulation, scheduling and operational follow-up across manufacturing campaigns.
- Developed a production calculator to automate ingredient conversion calculations, reducing manual work and saving operational time.

Product Development Technologist - Nutritionals

Apr 2018 - Jan 2020

Bega Cheese

- Developed formulations, bills of materials, specifications, operating conditions and regulatory documentation across nutritional powder categories.
- Used planning and business systems including M3 to support product development, formulation changes and manufacturing readiness.
- Built a goat milk formulation calculator in VBA to automate manufacturing operation calculations and improve process consistency.

Dry Blending Formulator and QA Officer

Sep 2017 - Apr 2018

Australian Natural Milk Association

- Supported dry-blending formulation, GMP audits, supplier documentation, quality systems, corrective actions and product-compliance reporting.

HEALTHCARE EXPERIENCE

Clinical Dietitian Placement

2016

Sydney Hospitals

- Completed clinical nutrition placements at Concord Hospital, Sydney Prince Alfred Hospital, Bondi Mental Health Clinic and Sydney Westmead Children's Hospital.
- Used electronic patient record and hospital information systems to review clinical notes, nutrition history and care documentation.
- Supported paediatric/infant nutrition, allergies and intolerances, rehabilitation nutrition, food-safety auditing and clinical research activities.

EDUCATION

- **Master of Computer Science, Artificial Intelligence specialisation** | Monash University | Completed Jun 2026

First in class Machine Learning; Computer Vision; Applied Practice 1.

Result High Distinction average.

- **Master of Science, Nutrition and Dietetics** | The University of Sydney | 2016
- **Bachelor of Science, Food Science and Nutrition** | The University of Auckland | 2014

First in class Food Quality Attribute (highest score).

Award Certificate of Distinction in Statistics for top 10%.

COURSEWORK EVIDENCE

- Relevant coursework: NLP, Machine Learning, AI, Computer Vision, Cryptography, Databases, Software Engineering, Python and Applied Practice.
- Databases and Data Handling: conceptual and logical database design, relational schema design, SQL scripting, data preprocessing and exploratory analysis.
- Systems and Networking: routing, EIGRP metric analysis, static routes, firewall configuration, Wireshark packet analysis and TCP/IP concepts.